SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Somaiya TRUST

FitTrack: Personal Health, Nutrition and Fitness Tracker

Web Programming Laboratory (WPL) Mini Project

Theme: SDG 3 - Good Health and Well-Being

Neel Parikh | Roll No. 16014024052

Shourya Ranjan | Roll No. 16014024057

Batch A-3 | Branch: EXCP

K. J. Somaiya College of Engineering, Mumbai

Agenda

- 1. Project Title and Team Details
- 2. Problem Statement
- 3. System Architecture
- 4. Features of the Designed System
- 5. Results and Demonstration Outcomes
- 6. Conclusion
- 7. References

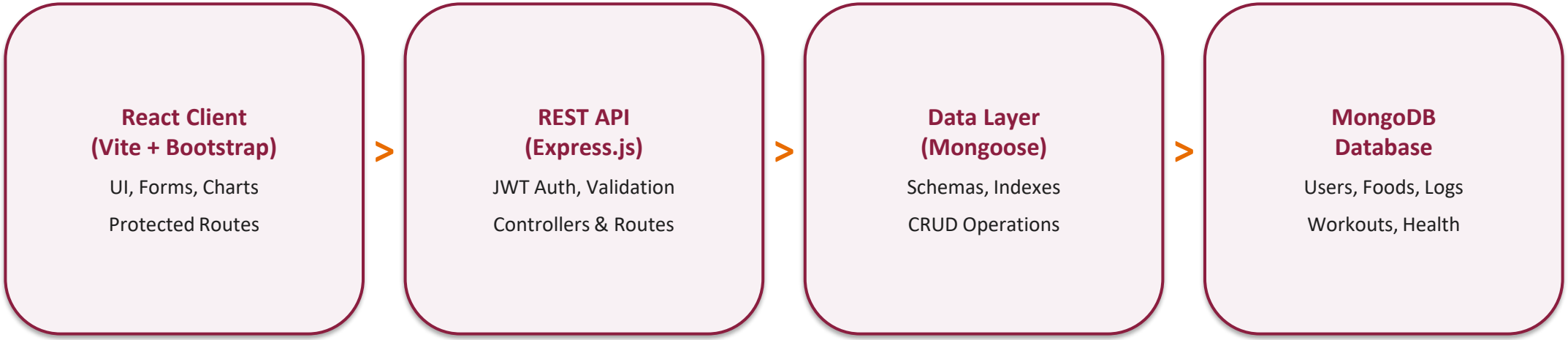
Problem Statement

- Many individuals struggle to consistently track nutrition, workouts and daily health habits in one place.
- Existing tools are often fragmented, complex, or lack an educational, student-friendly implementation.
- There is a need for a unified web application that helps users:
  - Monitor calories, protein and meals
  - Record strength and cardio workouts
  - Track water, sleep and weight
  - Visualize daily progress through dashboards and charts
- FitTrack addresses this need as a complete MERN-stack solution aligned with SDG 3: Good Health and Well-Being.
- Note: FitTrack is an educational tracking tool and does not provide medical diagnosis.

Project Objectives & SDG Alignment

- **Objectives**
  - Build a secure user authentication system using JWT
  - Enable nutrition search, logging and daily targets
  - Support workout creation with exercise library
  - Provide health habit tracking and progress charts
  - Demonstrate modular MERN architecture for viva
- **SDG 3 Contribution**
  - Promote awareness of daily nutrition choices
  - Encourage consistent physical activity tracking
  - Support hydration, sleep and weight monitoring
  - Help users build healthier everyday habits
  - Educational focus - estimates, not medical advice

System Architecture (MERN Stack)



Data Flow: Browser > Axios REST Calls > Express Middleware (JWT) > Mongoose Models > MongoDB Collections

Modular Backend & Frontend Design

- **Backend Modules (completely modular with clear function separation)**
  - Models: User, Food, FoodLog, Exercise, Workout, HealthLog
  - Controllers: Auth, Users, Foods, FoodLogs, Workouts, Health, Dashboard, Progress
  - Middleware: JWT protect, express-validator, centralized error handler
  - Utilities: nutrition target calculator (Mifflin-St Jeor), token generation
- **Frontend Modules**
  - Pages, reusable components, AuthContext, Axios service layer, layouts
  - Protected routes for dashboard pages; public routes for Home/About/Auth
- **This modular structure makes the project viva-friendly and maintainable.**

Technology Stack

- **Frontend**
  - React + Vite (JavaScript)
  - Bootstrap 5 + Bootstrap Icons
  - React Router, Axios
  - Chart.js / react-chartjs-2
  - Responsive custom CSS theme
- **Backend & Database**
  - Node.js + Express.js REST API
  - MongoDB + Mongoose ODM
  - JWT Authentication
  - bcryptjs password hashing
  - express-validator + CORS + dotenv

Features of the Designed System

Pillar 1

Nutrition & Protein Tracking

Food DB, meal logs, calorie & protein targets

Pillar 2

Workout & Exercise Tracking

Exercise library, sets/reps/cardio logs

Pillar 3

Health Dashboard & Progress

Water, sleep, weight, Chart.js analytics

Feature Deep Dive - Nutrition Module

- Seeded food database with categories: Protein, Grains, Fruits, Vegetables, Snacks
- Search + category filters; select food, quantity and meal type
- Dynamic nutrition scaling (example: Chicken Breast 150 g -> 247.5 kcal, 46.5 g protein)
- Food logs grouped by Breakfast, Lunch, Dinner and Snacks
- Custom food creation with validation
- Daily calorie and protein targets using Mifflin-St Jeor BMR x activity x goal
- Progress bars comparing intake vs target on Nutrition page and Dashboard

Feature Deep Dive - Workouts & Health

- **Workout Module**
  - Exercise library with search & filters
  - Categories: Strength, Cardio, Bodyweight, Flexibility
  - Log sets, reps, weight or duration/distance
  - Save completed workouts and view history
  - Delete and review workout details
- **Health & Progress**
  - Daily water intake (ml)
  - Sleep duration (hours)
  - Body weight logging
  - Dashboard summary cards
  - Charts for 7 / 30 / 90 days (real data only)

Authentication, Security & Validation

- User registration and login with email uniqueness checks
- Passwords hashed using bcryptjs (never stored in plaintext)
- JWT tokens issued on login/register; Authorization: Bearer middleware
- Protected frontend routes (/app/*) and protected REST endpoints
- Users can access only their own food logs, workouts and health data
- Input validation on both frontend forms and backend (express-validator)
- Centralized API error responses: { success, message/data }
- Secrets and Mongo URI stored in environment variables (.env)

Results - Working System Outcomes

- Successfully developed a complete, runnable MERN application: FitTrack
- Verified end-to-end flows:
  - User registration, login and profile target calculation
  - Food search, meal logging and daily nutrition totals
  - Workout creation from exercise library
  - Health habit logging and dashboard aggregation
  - Progress charts generated from real MongoDB data
- Seed data: 34 foods and 22 exercises for immediate usability
- Responsive UI with professional health/fitness visual design
- Application runs locally: Client :5173 | API :5000 | MongoDB :27017

Testing - Sample Test Cases

- TC01 Register with valid details -> Account created, JWT returned
- TC02 Register with duplicate email -> 400 validation error
- TC03 Login with wrong password -> 401 Unauthorized
- TC04 Access /api/dashboard without token -> 401 blocked
- TC05 Log Chicken Breast 150 g -> Calories 247.5, Protein 46.5 computed correctly
- TC06 Create workout with sets/reps -> Saved and visible in history
- TC07 Update profile age/height/weight -> Calorie & protein targets recalculated
- TC08 Progress page with no prior data -> Empty state (no fabricated charts)
- TC09 Mobile/responsive layout -> Sidebar collapses; no horizontal scroll

Conclusion

- FitTrack successfully demonstrates a full-stack MERN solution for personal health tracking.
- The system integrates three core pillars - Nutrition, Workouts and Health Progress - into one cohesive application.
- Modular architecture, JWT security, validation and real MongoDB persistence make the project robust and viva-ready.
- Aligned with SDG 3 by promoting awareness of everyday well-being habits through practical digital tracking.
- Future scope: meal templates, workout plans, PDF reports and enhanced analytics.
- Limitation: nutritional calculations are educational estimates and not medical advice.

References

- United Nations - Sustainable Development Goal 3: Good Health and Well-Being. https://sdgs.un.org/goals/goal3
- MongoDB Documentation. https://www.mongodb.com/docs/
- Express.js Documentation. https://expressjs.com/
- React Documentation. https://react.dev/
- Mongoose ODM Documentation. https://mongoosejs.com/docs/
- Chart.js Documentation. https://www.chartjs.org/docs/
- Mifflin, M. D. et al. - A new predictive equation for resting energy expenditure (Mifflin-St Jeor).
- Bootstrap 5 Documentation. https://getbootstrap.com/docs/5.3/
- JSON Web Token (RFC 7519) and bcrypt password hashing references.

SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Somaiya TRUST

Thank You

Questions & Discussion Welcome

Neel Parikh - 16014024052

Shourya Ranjan - 16014024057

Batch A-3 | EXCP | WPL Mini Project

K. J. Somaiya College of Engineering

Somaiya Vidyavihar University